

Optimizing Healthcare Staffing with AI-Driven Predictive Forecasting

A leading American healthcare provider struggled with staffing, patient demand, and operational efficiency. They partnered with Artha Solutions to implement an AI/ML predictive forecasting model. This solution helped them optimize staffing, reduce inefficiencies, and improve patient experience by accurately predicting demand.

Outcomes

35%

Optimized resource allocation saved staffing costs by 35% while maintaining efficiency.

40%

AI-driven forecasts aligned staff schedules with patient demand to improve scheduling accuracy by 40%.

25%

Optimized staff availability reduced patient wait time by 25%, thus improving patient experience.

99.5%

Predictive insights improved staffing forecasting decisions by 99.5% across 100+ clinics.



Challenges

Staffing Efficiency & Cost Management

The healthcare provider needed an intelligent staffing model to ensure clinics were neither understaffed nor overstaffed, preventing patient dissatisfaction and reducing unnecessary labor costs.

Data Gaps Across Clinics

With over 100 locations, some clinics had mature datasets (3+ years of data) while newly acquired clinics had limited historical data, making it difficult to create a unified forecasting model.

Multi-Seasonality & Data Variability

Clinic traffic exhibited daily, weekly, and seasonal fluctuations, making traditional forecasting models like ARIMA and Holt-Winters ineffective in predicting patient volume accurately.



Solutions Provided



Data Preprocessing & Discovery

Consolidated historical clinic data, cleaned missing values, and created a comprehensive time-series dataset, identifying multi-seasonality trends and structural breaks to enhance model accuracy.



Model Development & Implementation

Evaluated multiple forecasting models, selecting PROPHET for its ability to handle multi-seasonality, trend shifts, and irregular data, integrating CDC flu data to enhance prediction accuracy.



Predictive Insights & Optimization

Generated daily patient volume forecasts per clinic, enabling proactive staffing adjustments with real-time insights, optimizing workforce allocation, reducing wait times, and enhancing patient satisfaction.